

Comparing DVD Audio and Compact Disc

Specification	DVD-Audio	CD
Audio Format	PCM	PCM
Disk Capacity	4.7Gb - Single layer	650Mb
	8.5Gb - Dual Layer	
	17Gb - Double Sided Dual Layer	
Channels	Up to 6	2 (stereo)
Frequency Response	0 - 96kHz (max)	5 - 20kHz
Dynamic Range	144db	96db
Sampling Rate - 2 channel	44.1, 88.2, 176.4KHz or 48, 96, 192KHz	44.1kHz
Sampling Rate/Multi-channel	44.1, 88.2KHz or 48, 96KHz	n/a
Sample Size (Quantization)	12, 16, 20, or 24 bits	16 bits
Maximum Data Rate	9.6 Mbps.	1.4Mbps

The above table outlines the technical specifications for PCM on DVD-Audio and standard CD's.

Another factor is that for compatibility with DVD-Video players, DVD-Audio discs can also contain audio encoded using the Dolby Digital and/or DTS formats. Here is the catch. Say, you have a DVD-Audio disc which contains both PCM audio tracks and the same tracks encoded in either Dolby Digital or DTS. This disc, if inserted in a DVD-Audio player will playback the PCM tracks ONLY and ignores the encoded Dolby Digital or DTS tracks. This will happen even if the DVD-Audio player supports the DVD-Video standard. On the other hand, a DVD-Video player will only play the encoded Dolby Digital or DTS tracks and ignore the PCM tracks.

DVD-Audio discs can store PCM tracks as well as associate video and images to tracks making it an all-round experience for the end-user. Another factor that makes DVD-Audio discs